

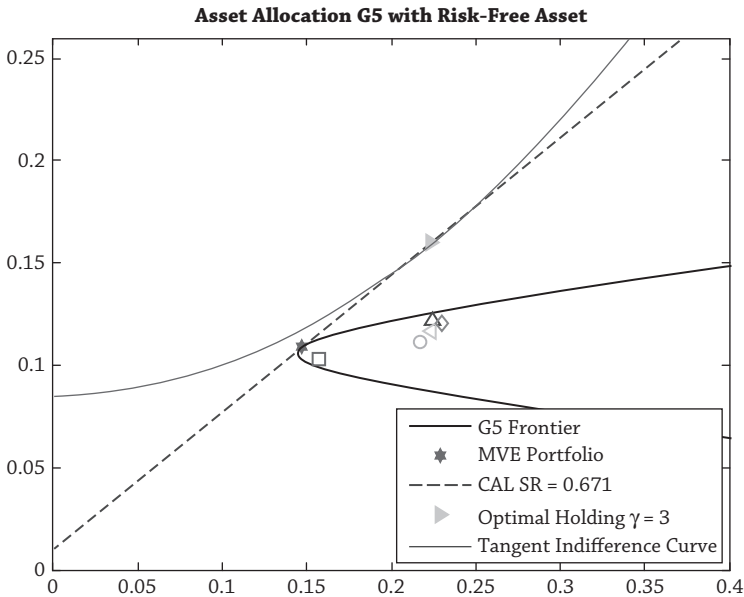


Figure 3.11

Finding the optimal combination of the MVE with the risk-free asset is equivalent to finding the point at which the highest possible indifference curve touches the CAL. The tangency point is the investor’s optimal portfolio. In Figure 3.11, we graph the CAL and show the optimal holding in the triangle for an investor with a risk aversion of $\gamma = 3$. The indifference curve that is tangent to this point—which corresponds to the maximum utility for this investor—is also plotted.

In Figure 3.11, the tangency point of the highest indifference curve and the CAL lies to the right of the MVE. This means the investor shorts the risk-free asset, or borrows money at 1%, and has a levered position in the MVE. The optimal positions corresponding to the triangle, which is the tangency MVE point, are:

	U.S.	JP	U.K.	GR	FR	Risk-Free
MVE Portfolio	0.80	0.37	0.18	0.15	0.03	-0.52

The proportions of the risky assets relative to each other in this optimal portfolio are the same as the weights of the MVE. That is, the 0.53 MVE weight of the United States is the same as $0.80 / (0.80 + 0.37 + 0.18 + 0.15 + 0.03)$. The optimal position for the $\gamma = 3$ investor has a Sharpe ratio of 0.671, which is the same as the CAL as it lies on the CAL.

How much has the investor gained in moving from our previous constrained setting in section 3.1 (no risk-free asset available) to the example with the risk-free asset included? The certainty equivalent of the tangency position with the